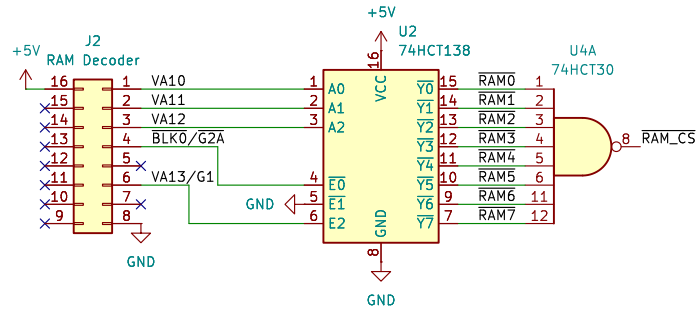
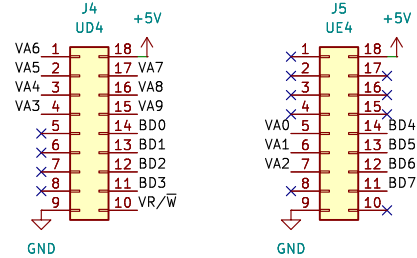


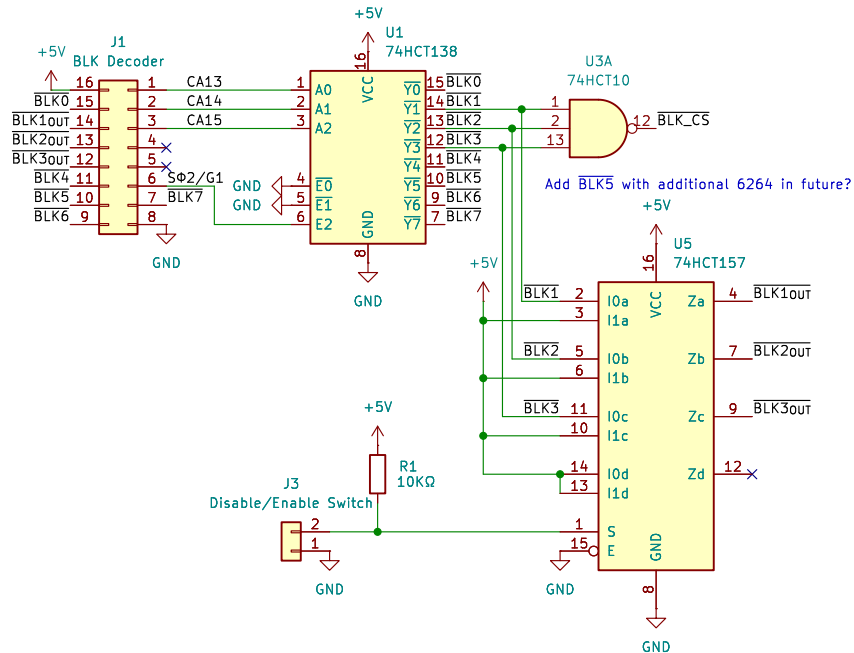
Signals to/from RAM decoder (74LS138)
Original decoder activates internal RAM only for RAM0 and RAM4 to RAM5
We want to activate our replacement for every 1KB chunk in BLK0, including RAM1 to RAM3
This will give us the full 8KB in BLK0 – standard 5KB plus 3KB RAM expansion



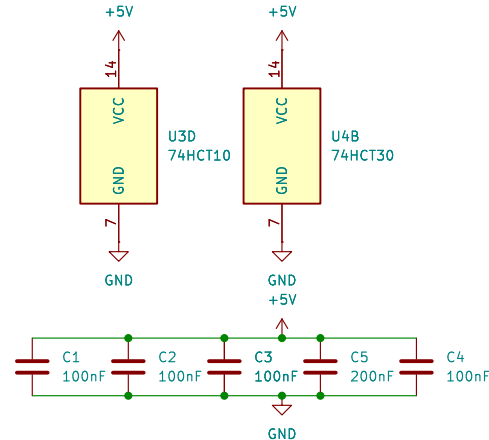
Original 2114 RAM chips are removed from the VIC.
Two are replaced with IC sockets so daughterboard can be installed.



Signals to/from BLK decoder (74LS138)
We want to pass BLK0 back so it can be used for BLK0 handling like normal (input to RAM decoder)
We want to capture BLK1 to BLK3 selects and use for our RAM expansion
But we also want to be able to disable the expansion for software compatibility
This can be done via a switch:
* When the switch is high, the BLK1/BLK2/BLK3 are passed back to the VIC, disconnecting our RAM expansion from the data bus and passing through to the cartridge port
* When the switch is low, these are blocked which prevents the VIC from disconnecting our RAM from the data bus

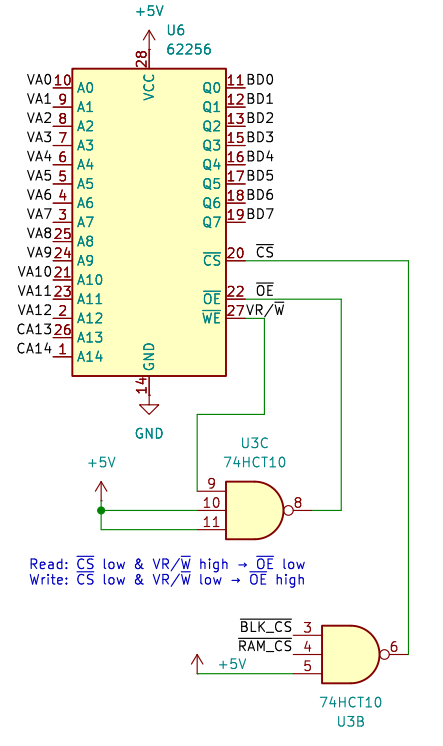


Power & Decoupling



Refer to the "DIP-16 Breakout Board" for the J1 & J2 connection

32KB RAM Expansion



Read: $\overline{CS}$ low & VR/ $\overline{W}$ high $\rightarrow$ $\overline{OE}$ low
Write: $\overline{CS}$ low & VR/ $\overline{W}$ low $\rightarrow$ $\overline{OE}$ high

U1	74HCT138 3-to-8 Decoder
U2	74HCT138 3-to-8 Decoder
U3	74HCT10 Triple 3-Input NAND
U4	74HCT30 8-Input NAND
U5	74HCT157 Quad 2-Input Multiplexer
U6	62256 32KB SRAM
R1	10K Ω Pullup Resistor
C1 TO C4	100nF Decoupling Capacitor
C5	200nF Decoupling Capacitor
J1 & J2	2x8 Dupont Female-Female Cables to VIC
J3	Wiring to external SPST switch
J4 & J5	Pair of 1x9 SIL Pin Headers or 2x9 Wirewrap Sockets

www.youtube.com/@Brfff

Brett Hallen

Sheet: /

File: VIC20_Internal_32KB_RAM.kicad_sch

Title: Commodore VIC-20 32KB RAM Expansion

Size: A4

Date: 25/MAY/2026

Rev: A

KiCad E.D.A. 9.0.9

Id: 1/1